S. N.	Topic	No.
1.	Multidisciplinary nature of environmental studies Definition, scope and importance.	1
2.	**Natural Resources:** Renewable and non-renewable resources, Natural resources and associated problems.	1
3.	**a) Forest resources:** Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people.	1
4.	**b) Water resources:** Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems.	1
5.	**c) Mineral resources:** Use and exploitation, environmental effects of extracting and using mineral resources, case studies.	1
6.	**d) Food resources:** World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies.	1
7.	**e) Energy resources:** Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources. Case studies.	1
8.	**f) Land resources:** Land as a resource, land degradation, man induced landslides, soil erosion and desertification.	1
9.	Role of an individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles.	1
10.	**Ecosystems**: Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem	1
11.	Ecological succession, Food chains, food webs and ecological pyramids. Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)	2
12.	**Biodiversity and its conservation: -** Introduction, definition, genetic, species & ecosystem diversity and bio geographical classification of India.	1
13	**Value of biodiversity:** consumptive use, productive use, social, ethical, aesthetic and option values.	1
14.	Biodiversity at global, National and local levels, India as a mega-diversity nation. Hot-sports of biodiversity.	1
15.	**Threats to biodiversity:** habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India. Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.	1
16.	**Environmental Pollution**: definition, cause, effects and control measures of: a. Air pollution b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards.	1

17.	**Solid Waste Management**: causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution.	1
18.	**Social Issues and the Environment**: From Unsustainable to Sustainable development, Urban problems related to energy, Water conservation, rain water harvesting, watershed management.	1
19.	**Environmental ethics:** Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. dies.	1
20.	Wasteland reclamation. Consumerism and waste products.	1
21.	Environment Protection Act. Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Issues involved in enforcement of environmental legislation. Public awareness.	1
22.	**Human Population and the Environment:** population growth, variation among nations, population explosion, Family Welfare Programme.	1
23.	**Environment and human health**: Human Rights, Value Education, HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health.	1
24.	**Natural Disasters**- Meaning and nature of natural disasters, their types and effects. Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves, Climatic change: global warming, Sea level rise, ozone depletion.	3
25.	**Man Made Disasters-** Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, air pollution, water pollution, deforestation, industrial waste water pollution, road accidents, rail accidents, air accidents, sea accidents.	2
26.	**Disaster Management-** Effect to migrate natural disaster at national and global levels. International strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community –based organizations and media. Central, state, district and local administration; Armed forces in disaster response; Disaster response; Police and other organizations.	3

INTRODUCTION

The word environment is derived from the French word '**environner**' meaning surroundings. Hence, everything surrounding us is called "**ENVIRONMENT**".

Environment literally means Surrounding in which we are living. Environment includes all those things on which we are directly or indirectly dependent for our survival, whether it is living component like animals, plants or non living component like soil, air water.

Environmental science is essentially the application of scientific methods and principles to the study of environmental issues, so it has probably been around in some forms as long as science itself. Environmental science is often confused with other fields of related interest, especially ecology, environmental studies, environmental education and environmental engineering. Environmental science is not constrained with any one discipline and it is a comprehensive field.

There are two types of environments:
1. Natural environment
2. Man made environment

1.Natural: The environment in its original form without the interference of human beings is known as natural environment. It operates through self regulating mechanism called homeostasis *i.e*, any change in the natural ecosystem brought about by natural processes is counter balanced by changes in other components of environment.

2.Man made or Anthropogenic Environment: The environment changed or modified by the interference of human beings is called man made environment. Man is the most evolved creature on this earth. He is modifying the environment according to his requirements without bothering for its consequences. Increased technologies and population explosion are deteriorating the environment more and more.

Meaning Of Environmental Studies:
Environmental studies are the scientific study of the environmental system and the status of its inherent or induced changes on organisms. It includes not only the study of physical and biological characters of the environment but also the social and cultural factors and the impact of man on environment.

Objectives and Guiding Principles of Environmental Studies:
According to UNESCO (1971), the objectives of environmental studies are:

(a) Creating the awareness about environmental problems among people.

(b) Imparting basic knowledge about the environment and its allied problems.

(c) Developing an attitude of concern for the environment.

(d) Motivating public to participate in environment protection and environment improvement.

(e) Acquiring skills to help the concerned individuals in identifying and solving environmental problems.

(f) Striving to attain harmony with Nature.

According to UNESCO, the guiding principles of environmental education should be as follows:

(a) Environmental education should be compulsory, right from the primary up to the post graduate stage.

(b) Environmental education should have an interdisciplinary approach by including physical, chemical, biological as well as socio-cultural aspects of the environment. It should build a bridge between biology and technology.

(c) Environmental education should take into account the historical perspective, the current and the potential historical issues.

(d) Environmental education should emphasise the importance of sustainable development i.e., economic development without degrading the environment.

(e) Environmental education should emphasise the necessity of seeking international cooperation in environmental planning.

(f) Environmental education should lay more stress on practical activities and first hand experiences.

Scope and Importance of Environmental Studies:

The disciplines included in environmental education are environmental sciences, environmental engineering and environmental management.

(a) Environmental Science:

It deals with the scientific study of environmental system (air, water, soil and land), the inherent or induced changes on organisms and the environmental damages incurred as a result of human interaction with the environment.

(b) Environmental Engineering:

It deals with the study of technical processes involved in the protection of environment from the potentially deleterious effects of human activity and improving the environmental quality for the health and well beings of humans.

(c) Environmental Management:
It promotes due regard for physical, social and economic environment of the enterprise or projects. It encourages planned investment at the start of the production chain rather than forced investment in cleaning up at the end.

It generally covers the areas as environment and enterprise objectives, scope, and structure of the environment, interaction of nature, society and the enterprise, environment impact assessment, economics of pollution, prevention, environmental management standards etc.

The importance's of environmental studies are as follows:

1. To clarify modern environmental concept like how to conserve biodiversity.

2. To know the more sustainable way of living.

3. To use natural resources more efficiently.

4. To know the behaviour of organism under natural conditions.

5. To know the interrelationship between organisms in populations and communities.

6. To aware and educate people regarding environmental issues and problems at local, national and international levels.

Need of Public Awareness about Environment:
In today's world because of industrialization and increasing population, the natural resources has been rapidly utilised and our environment is being increasingly degraded by human activities, so we need to protect the environment.

It is not only the duty of government but also the people to take active role for protecting the environment, so protecting our environment is economically more viable than cleaning it up once, it is damaged.

The role of mass media such as newspapers, radio, television, etc is also very important to make people aware regarding environment. There are various institutions, which are playing positive role towards environment to make people aware regarding environment like BSI (Botanical Survey of India, 1890), ZSI (Zoological Survey of India, 1916), WII (Wild Life Institute of India, 1982) etc.

Institutions in environment

1.The Bombay Natural History Society (BNHS), :Mumbai: It was founded on 15 September 1883, is one of the largest non-governmental organizations in India engaged in conservation and biodiversity research.

2. World Wide fund for nature- India (WWF-1), New Delhi: The WWF-1 was initiated in 1969 in Mumbai, after which the headquaters were shifted to Delhi with several State.

3.Centre or science and environment (CSE), New Delhi: is a public interest research and advocacy organisation based in New Delhi.

4.C.P.R Environmental Education Centre, Madras: the CPR-EEC was set up in 1988 CPREEC) is a Centre of Excellence of the Ministry of Environment and Forests (MoEF)

5.The **Centre for Environment Education** (CEE) in India was established in August 1984 as a Centre of Excellence supported by the Ministry of Environment and Forests.

6.Bharati Vidyapeeth University, Institute of Environment Education & Research, Pune was established in 1993.This is part of the Bharati Vidyapeeth deemed University.

7. **The Salim Ali Center for Ornithology and Natural History (SACON)**: **-** It is an autonomous organization with headquarters at Coimbatore.

8.Wild life Institute of India (WII), Dehradhun: Is an autonomous institution of MoEF, GOI, established in 1982.

9.Zoological survey of India (ZSI): is a premier organisation in zoological research and studies.

10.The madras Crocodile Bank Trust (MCBT): MCBT, the first crocodile conservation breeding in Asia, was founded in 1976.

11.Uttarkhand seva nidhi (USKN), Almora : It is a public charitable trust founded in 1967.

12.The **Botanical Survey of India (BSI)** is an institution set up by the Government of India in 1887 to survey the plant resources of the Indian empire.

People in environment

Charles Darwin: wrote the origin of species, which brought to light the close relationship between habitats and species. It brought about a new way of thinking about man's relationship with other species that was based on evolution.

Ralph Emerson: spoke of the dangers of commerce to our environment way back in the 1840s.

Henry Thoreau: in the 1860s wrote that the wilderness should be preserved after he had lived in the wilderness for a year.

John Muir: He was a Scottish-born American naturalist, author, and early advocate of preservation of wilderness in the United States.

Aldo Leopold: was a forest official in the US in the 1920s. He designed the early policies on wilderness conservation and wildlife management. He was considered the father of wildlife ecology and a true Wisconsin hero. His book, '*A Sand County Almanac'* is acclaimed as the century's literary landmark in conservation, which guided many to 'live in harmony with the land and with one another'.

Rachel Carson : was an American marine biologist and conservationist whose writings are credited with advancing the global environmental movement. She was nature writer, and some of the books like '*The Sea Around Us' and 'The Edge of the Sea*' are to her credit.

EO Wilson: is an entomologist who envisioned that biological diversity was a key to human survival on Earth. He wrote 'Diversity of life' in 1993, which was awarded a prize for the best book published on environmental issues.

Salim Ali: was an Indian ornithologist and naturalist, Known as the "birdman of India",

Smt. Indira Gandhi: as PM played a very significant role in the preservation of India's wildlife.

S P Godrej was one of Inida's greatest supports of wildlife conservation n and nature awareness programs.

M. S. Swaminathan: He has founded the MS Swaminathan Research Foundation in Chennai, which does work on the conservation of biological diversity.

Madhav Gadgil is a well-known ecologist in India. His interests range from board ecological issues such as developing community Biodiversity Registers and conserving sacred groves to studies on the behaviour of mammals, birds and insects.

M. C. Mehta: Environmental lawyer. Initiated the Government to implement Environmental education in schools and colleges, struggles for protection of Taj Mahal and cleaning of Ganga water.

Anil Agarwal : a journalist who wrote the first report on the state of India's Environment in 1982. He was the founder of CES, an active NGO that supports various environmental issues.

Medha Patkar: known as one of rural India's champions, has supported the cause of the downtrodden tribal people whose environment is being affected by the dams on the Narmada river.

Lect.2.Natural Resources: Renewable and non-renewable resources, Natural resources and associated problems.

Natural resources can be defined as 'variety of goods and services provided by nature which are necessary for our day-to-day lives'. Eg: Plants, animals and microbes (living or biotic part), Air, water, soil, minerals, climate and solar energy (non- living or abiotic part). They are essential for the fulfillment of physiological, social, economical and cultural needs at the individual and community levels

Life on this planet earth depends upon a variety of goods and services provided by the nature, which are known as Natural resources. Thus water, air, soil, minerals, coal, forests, crops and wildlife are all examples of natural resources. Any stock or reserve that can be drawn from nature is a natural resource.

The natural resources are two kinds:

i) Renewable resources: which are in exhaustive and can be regenerated within a given span of time.
Example- forests, wildlife, wind energy, biomass energy, tidal energy, hydro power. Solar energy is also are renewable energy form of energy as it is an in-exhaustible source of energy.

(ii) **Non-renewable resources**: which cannot be regenerated. Example- Fossil fuels like coal, petroleum, minerals etc. Once we exhaust these reserves, the same cannot be replenished. Even our renewable resources can become non-renewable if we exploit them to such extent that the irrate of consumption exceeds their rate of regeneration. It is very important to protect and conserve our natural resources and use them in a judicious manner so that we do not exhaust them. It does not mean that we should stop using most of the natural resources, but we should use the resources in such a way that we always save enough of them for our future generations.

Major natural resources:-Forest resources Water resources Mineral resources Food resources Energy resources Land resources

S.No	Renewable energy resources	Non-renewable energy resources
1.	It can be used again and again throughout its life.	It cannot be used again and again but one day it will be exhausted.
2.	These are the energy resources which cannot be exhausted.	They are the energy resources which can be exhausted one day.
3.	It has low carbon emission and hence environment friendly.	It has high carbon emission and hence not environment friendly.
4.	It is present in unlimited quantity.	It is present in limited quantity and vanishes one day
5.	Cost is low.	Cost is high.
6.	Renewable energy resources are pollution free.	The non-renewable energy resources are not pollution free.
7.	Life of resources is infinite.	Life of resources is finite and vanishes one day.
8.	It has high maintenance cost.	It has low maintenance cost as compared with the renewable energy resources.
9.	Large land area is required for the installation of its power plant.	Less land area is required for its power plant binstallation.
10.	Solar energy, wind energy, tidal energy etc are the examples of renewable resources.	Coal, petroleum, natural gases are the examples of non-renewable resources

Lect.3. a)* Forest resources*: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people.

A forest can be defined as a biotic community predominant of trees, shrubs or any other woody vegetation usually in a closed canopy. It is derived from latin word *'foris'* means *'outside'*.

Forest are one of the most important natural resources on this earth. Covering the earth like a green blanket these forests not only produce in numberable material goods but also provide several environmental services which are essential for life.

India's Forest Cover is 6,76,000 sq.km (20.55% of geographic area). Scientists estimate that India should ideally have 33% of its land under forests. Today we only have about 12% thus we need not only to protect our existing forests but also to increase our forest cover.

Use and overexploitation:

Scientists estimate that India should ideally have 33 percent of its land under forests. Today we have only about 12 percent. Thus we need not only to protect existing forests but also to increase our forest cover.

People who live in or near forests know the value of forest resources first hand because their lives and livelihoods depend directly on these resources.

However, the rest of us also derive great benefits from the forests which we are rarely aware of. The water we use depends on the existence of forests on the watersheds around river valleys. Our homes, furniture and paper are made from wood from the forest. We use many medicines that are based on forest produce. And we depend on the oxygen that plants give out and the removal of carbon dioxide we breathe out from the air.

Forests once extended over large tracts of our country. People have used forests in our country for thousands of years. As agriculture spread the forests were left in patches which were controlled
mostly by tribal people. They hunted animals and gathered plants and lived entirely on forest resources. Deforestation became a major concern in British times when a large amount of timber was extracted for building their ships.

This led the British to develop scientific forestry in India. They however alienated local people by creating Reserved and Protected Forests which curtailed access to the resources. This led to a loss of stake in the conservation of the forests which led to a gradual degradation and fragmentation of forests across the length and breadth of the country.

Forest Functions :
I. Protective and ameliorative functions.
II. Productive functions
III. Recreational and educational functions
IV. Development functions

A. Watershed protection

1. Reducing the rate of surface run-off of water
2. Preventing flash floods and soil erosion
3. Producing prolonged gradual run-off and thus safeguarding against drought.

B. Erosion control

Holding soil (by preventing rain from from directly washing soil away)

C. Land bank

Maintaining soil nutrients and structure.

D. Atmospheric regulation

- Absorption of solar heat during evapotranspiration
- Maintaining carbon dioxide levels for plant growth
- Maintaining the local climatic conditions

II. Productive Functions

Local use – Consumption of forest produce by local people who collect it for sustenance

Food: (comsumptive use) gathering plants, fishing, hunting from the forest.
Fodder for cattle
Fuel wood and charcoal for cooking and heating
Poles for building homes in rural and wilderness areas
Timber for house hold articles and construction
Fiber for weaving baskets, ropes, nets, strings, etc.,

Sericulture for silk

Apiculture for rearing bees for honey (bees as pollinators) Medicinal plants for traditional medicines, investigating them as potential source for new modern drugs

Market use (productive use) Most of the products used for consumptive purposes and good source of income for supporting their livelihood of forest dwelling people.

Minor forest products (NTFPs): Fuel wood, fruits, gum, fiber, etc which are collected and solid in local markets as a source of income for forest dwellers

Major timber extraction for construction, industrial uses, paper pulp etc. Timber extraction is done in India by the forest department, but illegal logging continues in many of the forests of India and the world.

III. ***Recreational And Educational Functions***: Eco tourism

IV. ***Developmental Functions***
Employment functions
Revenue

Ecological significance of forests:

1. Balances CO_2 and O_2 levels in atmosphere.
2. Regulates earth temperature and hydrological cycle
3. Encourage seepage and reduces runoff losses, prevents drought
4. Reduces soil erosion (roots binding), prevents siltation and landslides thereby floods
5. Litter helps in maintaining soil fertility
6. Safe habitat for birds, wild animals and organisms against wind, solar radiation and rain

Deforestation:

Deforestation refers to the loss of forest cover; land that is permanently converted from forest to agricultural land, golf courses, cattle pasture, home, lakes or desert. The FAO (Food and Agriculture Organization of the UN) defines tropical deforestation as " change of forest with depletion of tree crown cover more than 90%" depletion of forest tree crown cover less than 90% is considered forest degradation

.Causes for Deforestation:

1. Agriculture: Conversion of forests to agricultural land to feed growing numbers of people
2. Commercial logging: (which supplies the world market with woods such as meranti, teak, mahogany and ebony) destroys trees as well as opening up forest for agriculture. Cutting of trees for fire wood and building material, the heavy lopping of foliage for fodder and heavy grazing of saplings by domestic animals like goals.
3. The cash crop economy: Raising cash crops for increased economy.
4. Mining
5. Increase in population: The needs also increase and utilize forests resources.
6. Urbanization & industrialization
7. Mineral exploration
8. Construction of dam reservoirs
9. Infrastructure development
10. Forest fires
11. Human encroachment & exploitation
12. Pollution due to acid rain

Environmental effects //Consequences of deforestation

1. Food problems
2. Ecological imbalance
3. Increasing CO_2
4. Floods leading to soil erosion
5. Destruction of resources
6. Heavy siltation of dams
7. Changes in the microclimate
8. Loss of biodiversity
9. Dessication of previously moist forest soil
10. Heavy rainfall and high sunlight quickly damage the topsoil in clearings of the tropical rainforests. In such circumstance, the forest will take much longer to regenerate and the land will not be suitable for agricultural use for quite some time.
11. Where forests are replanted, their replacement can mean a loss of quality

12. Loss of future markets for ecotourism. The value of a forest is often higher when it is left standing than it could be worth when it is harvested.
13. Some indigenous peoples' way of life and survival are threatened by the loss of forests. Fewer trees results an insecure future for forest workers
14. Deforestation can cause the climate to become extreme in nature. The occurrence and strength of floods and droughts affecting the economy.
15. The stress of environmental change may make some species more susceptible to the effect of insects, pollution, disease and fire
16. Most humid regions changes to desert
17. Environmental pollution
18. Global warming

Conservation Conservation derived from two Latin words, *con* – together,- *servare* – to keep or guard measures, *i.e.* an act of preservation or to keep together **.**

Concepts in conservation

1. Restraining cutting of trees and submerging the forests
2. Reforestation
3. Afforestation
4. Control forest diseases and forest fire
5. Recycling forest products
6. Replacing forest products
7. Avoids diversion of forest lands for other activities through acts like Forest Conservation Act and Wild life (protection) Act
8. Bringing awareness among people ex**:** Chipko movement, Appiko , Narmada Bachao Andolan
9. Implementing people's participatory programmes. Ex: Joint Forestry Manangement (JFM)

TIMBER EXTRACTION

- Logging for valuable timber such as teak and mahogany not only involves a few large trees per hectare but about a dozen more trees since they are strongly interlocked with each other by vines etc.
- Also road construction for making approach to the trees causes further damage to the forests.
- In India, firewood demand would continue to rise in future mostly consumed in rural areas, where alternative sources of energy, are yet to reach.

MINING

- Mining is the process of removing deposits of ores from substantially very well below the ground level.
- Mining is carried out to remove several minerals including coal.
- These mineral deposits invariably found in the forest region, and any operation of mining will naturally affect the forests.
- Mining from shallow deposits is done by surface mining while that from deep deposits is done by sub-surface mining.

- More than 80,000 ha of land of the country are presently under the stress of mining activities.

Effects of mining resources

- Mining operation require removal of vegetation along with underlying soil mantle and overlying rock masses. This results in destruction of landscape in the area.
- Large scale of deforestation has been reported in Mussorie and Dehradun valley due to mining of various areas.
- Indiscriminate mining in Goa since 1961 has destroyed more than 50,000 ha of forest land.
- Mining of radioactive mineral in Kerala, Tamilnadu and Karnataka are posing similar threats of deforestation.

DAMS AND THEIR EFFECTS ON FORESTS AND TRIBAL PEOPLE

- Big dams and river valley projects have multi-purpose uses and have been referred to as "Temples of modern India".
- India has more than 1550 large dams, the maximum being in the state of Maharashtra (more than 600) followed by Gujarat (more than 250) and Madhya Pradesh (130).
- The highest one is Tehri dam, on river Bhagirathi in Uttaranchal and the largest in terms of capacity is Bhakra dam on river Sutlej.

Effects on Tribal people

- The greatest social cost of big dam is the widespread displacement of local people.
- It is estimated that the number of people affected directly or indirectly by all big irrigation projects in India over the past 50 years can be as high as 20 millions.

- The Hirakud dam, one of the largest dams executed in fifties, has displaced more than 20,000 people residing in 250 villages.

Effects on forests

Thousands of hectares of forests have been cleared for executing river valley projects which breaks the natural ecological balance of the region.

Floods, landslides become more prevalent in such areas. Eg: The Narmada sagar project alone has submerged 3.5 lakh hectares of best forest comprising of rich teak and bamboo forests. The Tehri dam submerged 1000 hectares of forest affecting about 430 species of plants according to the survey carried out by the botanical survey of India.

Lect.4. b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems.

Water claims to be an important resource. An important use of water in our country is for irrigation. Besides, water is also required in large amounts for industrial and domestic consumption. Water resources of a country constitute one of its vital assets.

India receives annual precipitation of about 4000 km. The rainfall in India shows very high spatial and temporal variability and paradox of the situation is that Mousinram near Cherrapunji, which receives the highest rainfall in the world, also suffers from a shortage of water during the non-rainy season, almost every year.

The total average annual flow per year for the Indian rivers is estimated as 1953 km. The total annual replenishable groundwater resources are assessed as 432 km.

Significant of Water

1. It is revealed by the history of human civilization that water supply and civilization are most synonymous.

2. Several cities and civilizations have disappeared due to water shortages originating from climatic changes.

3. Millions of people all over the world, particularly in the developing countries, are losing their lives every year from waterborne disease.

4. An understanding of water chemistry is the basis of knowledge of the multidimensional aspects of aquatic environment chemistry, which involve the sources, composition, reactions, and transport of the water.

5. About 97% of the earth's water supply is in the ocean, which is unfit of the remaining 3%, 2% is locked in the polar ice-caps and only 1% is available as fresh water in rivers, lakes, streams, reservoirs and ground water which is suitable for human consumption.

Main Sources of Water

Main sources of water for our use are:

1. ***Rainfall:*** India can be broadly divided into 15 ecological regions. The vast ecological diversity of this country is reflected in the diversity in available water resources. With an average annual rainfall of 1170 mm, India is one of the wettest countries in the world.

However, there are large variations in the seasonal and geographical distribution of rainfall over the country. At one extreme are areas like Cherrapunji, in the northeast, which is drenched each year with 11,000 mm of rainfall, and at the other extreme are places. like Jaisalmer, in the west, which receives barely 200 mm of annual rainfall.

Though the average rainfall is adequate, nearly three quarters of the rain pours down in less than 120 days, from June to September.

2. ***Groundwater:*** India's groundwater resources are almost ten times its annual rainfall. According to the Central Groundwater Board of the Government of India, the country has an annual exploitable groundwater potential of 26.5 million hectare-meters.

Nearly 85% of currently exploited groundwater is used only for irrigation. Groundwater accounts for as much as 70-80% of the value of farm produce attributable to irrigation. Besides, groundwater is now the source of four-fifths of the domestic water supply in rural areas, and around half that of urban and industrial areas.

However, according to the International Irrigation Management Institute (IIMI), the water table almost everywhere in India is falling at between one to three meters every year. Furthermore, the IIMI estimates that

India is using its underground water resources at least twice as fast they are being replenished. Already, excessive ground water mining has caused land subsidence in several regions of Central Uttar Pradesh.

3. ***Surface water:*** There are 14 major, 44 medium and 55 minor river basins in the country. The major river basins constitute about 83-84% of the total drainage area. This, along with the medium river basins, accounts for 91% of the country's total drainage. India has the largest irrigation infrastructure in the world, but the irrigation efficiencies are low, at around 35%.

CONFLICTS OVER WATER

Water is essential for our existence and is fast becoming scarce. Rapidly increasing population and limited water resources give rise to conflicts over water.

Conflict through use: Unequal distribution of water leads to inter-state or international disputes.

Examples:

International conflicts

1. Conflict over water from the Indus between India and Pakistan
2. Conflict over water from the Colorado river between Mexico and USA
3. Conflict over water from the Shatt-al-Arab between Iran and Iraq
4. Conflict over water from the Bhramaputra between India and Bangladesh

National conflicts

1. Sharing of Cauvery water between Karnataka and Tamilnadu
2. Sharing of Krishna water between Karnataka and Andhra Pradesh
3. Sharing of Siruveni water between Tamilnadu and Kerala

Construction of dams or power stations:
For hydroelectric power generation, dams are built across the rivers, and this initiates conflicts between the states.

Conflict through pollution:
Rivers are also used for industrial purposes. They act as reservoirs for supply of fresh water and also a receptor of waste water and rubbish from the industry. Water crossing borders that has been polluted by wastes from one country develops into an international conflict.

Management of water conflicts

- Concerted efforts are required to enforce laws that check these practices to control water pollution
- In order to overcome the problem of sharing river water in a country, the concept of interlinking of rivers has been suggested
- Rivers should be nationalized, the National Water Authority and River Basin Authority should be given powers to ensure equitable distribution of basin water.

Floods

Technically, flooding occurs when the water level in any stream, river, bay or lake rises above bank full. Bays may flood as the result of a tsunami or tidal wave induced by an earthquake or volcanic eruption; or as a result of a tidal storm surge caused by a hurricane or tropical storm moving inland. Streams, rivers and lakes may be flooded by high amounts of surface runoff resulting from widespread precipitation or rapid snowmelt.

On a smaller scale, flash floods due to extremely heavy precipitation occurring over a short period of time can flood streams, creeks and low lying areas in a matter of a few hours. Thus, there are various temporal and spatial scales of flooding. Historical evidence suggests that flooding causes greater loss of life and property than any other natural disaster.

The magnitude, seasonality, frequency, velocity, and load are all properties of flooding which are studied by meteorologists, climatologists and hydrologists.

Spring and winter floods occur with some frequency primarily in the mid-latitude regions of the earth, and particularly where continental climate is the norm. Five climatic features contribute to the spring and winter flooding potential of any individual year or region:

1. Heavy winter snow cover;
2. Saturated soils or soils at least near their field capacity for storing water;
3. Rapid melting of the winter's snow pack;
4. Frozen soil conditions which limit infiltration; and
5. Somewhat heavy rains, usually from large scale cyclonic storms.

Drought:

In most arid regions of the world the rains are unpredictable. This leads to periods when there is a serious scarcity of water to drink, use in farms, or provide for urban and industrial use.

Drought prone areas are thus faced with irregular periods of famine. Agriculturists have no income in these bad years, and as they have no steady income, they have a constant fear of droughts. India has 'Drought Prone Areas Development Programs', which are used in such areas to buffer the effects of droughts. Under these schemes, people are given wages in bad years to build roads, minor irrigation works and plantation programs.

Drought has been a major problem in our country especially in arid regions. It is an unpredictable climatic condition and occurs due to the failure of one or more monsoons. It varies in frequency in different parts of our country.

DAMS:

It can be unequivocally stated that dams have made significant contributions to human development and the benefits derived from them have been considerable. Large dams are designed to control floods and to help the drought prone areas, with supply of water.

But large dams have proved to cause catastrophic environmental damage. Hence an attempt has been made to construct small dams. Multiple small dams have less impact on the environment.

Benefits:

Dams ensure a year round supply of water for domestic use and provide extra water for agriculture, industries and hydropower generation.

Problems:

They alter river flows, change nature's flood control mechanisms such as wetlands and flood plains, and destroy the lives of local people and the habitats of wild plant and animal species, particularly is the case with mega dams.

Some of the problems are mentioned below.

- Dam construction and submersion leads to significant loss of areable farmland and forest and land submergence
- Siltation of reservoirs, water logging and salination in surrounding lands reduces agricultural productivity
- Serious impacts on ecosystems - significant and irreversible loss of species and ecosystems, deforestation and loss of biodiversity, affects aquaculture
- Socio economic problems for example, displacement, rehabilitation and resettlement of tribal people.
- Fragmentation and physical transformation of rivers
- Displacement of people - People living in the catchment area, lose property and livelihood
- Impacts on lives, livelihoods, cultures and spiritual existence of indigenous and tribal people
- Dislodging animal populations
- Disruption of fish movement and navigational activities
- Emission of green house gases due to rotting of vegetation
- Large landholders on the canals get the lion's share of water, while poor and small farmers get less and are seriously affected leading to conflicts. Irrigation to support cash crops like sugarcane produces an unequal distribution of water.
- Natural disasters – reservoirs induced seismicity, flash floods etc and biological hazards due to large-scale impounding of water – increase exposure to vectorbrone
- diseases, such as malaria, schistosomiasis, filariasis

lect.5 c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.

Minerals provide the material used to make most of the things of industrial- based society; roads, cars, computers, fertilizers, etc. Demand for minerals is increasing world wide as the population increases and the consumption demands of individual people increase. The mining of earth's natural resources is, therefore accelerating, and it has accompanying environmental consequences.

Use and Exploitation:

The use of minerals varies greatly between countries. The greatest use of minerals occurs in developed countries. Like other natural resources, mineral deposits are unevenly distributed around on the earth. Some countries are rich in mineral deposits and other countries have no deposits. The use of the mineral depends on its properties. For example aluminum is light but strong and durable so it is used for aircraft, shipping and car industries.

(a) Mining is hazardous occupation:

1. This occupation involves several health risk dust produced during mining operation are injurious to health and cause lung diseases.

2. Extraction of some toxic or radioactive minerals leads to life threatening hazards.

3. Dynamite explosion during mining is very risky as fumes produced are extremely poisonous.

4. Underground mining is more hazardous than surface mining as there are more chances if accidents like roof falls, flooding and inadequate ventilation etc.

(b) Rapid depletion of high grade minerals:

Increasing demand for high grade minerals has compelled miners to carry out more extraction of minerals, which require more energy sources and produce large amount of waste materials.

(c) Wastage of upper soil layer and vegetation:

Surface mining results in the complete destruction of upper soil layer and vegetation. After extraction, the wastes are dumped in an area which destroys the total surface and vegetation.

(d) Environmental problems:

Over exploitation of mineral resources resulted in many environmental problems like:

1. Conversion of productive land into mining and industrial areas.

2. Mining and extraction process are one of the sources of air, water and land pollution.

3. Mining involves huge consumption of energy resources like coal, petroleum, natural gas etc. which are in-turn non renewable sources of energy.

Lect.6. d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studi

World Food Day – October 16th   World Water Day- March 22nd

Our food comes almost entirely from agriculture, animal husbandry and fishing *i.e.,* - 76% from crop lands, 17% from range lands *i.e.*, meat from grazing livestock and 7% - marine and fresh water *i.e.*, fisheries. The FAO (Food & Agricultural Organization of UN) defines sustainable agriculture as the one which conserves land, water and plant and animal genetic resources, does not degrade the environment and is economically viable and socially acceptable.

The report, "The Food Gap –the Impacts of Climate Change on Food Production: A 2020 Perspective", produced after a year-long assessment by America's Universal Ecological Fund (FEU-US), revealed that :

- *Global food production would not meet the food requirements of the world's estimated 7.8 billion people by 2020.*
- *Food prices are expected to jump by 20% in the next ten years as prolonged droughts and floods take their toll on food production.*
- *The report, which looked at the impact of climate change on four cereals - wheat, rice, maize and soybean - pointed out that*

❖ *global wheat production will experience a 14 percent deficit between production and demand*
❖ *Rice production will experience 11 percent deficit, and*
❖ *9 percent deficit in maize production.*
❖ *Soybean is the only crop showing an increase in global production, with an estimated five percent surplus.*

World Food Problems and Environmental Concerns:

1) Population growth: Food production in 64 of the 105 developing countries is lagging behind their population growth levels.

2) Poor agricultural practices: Poor environmental agricultural practices such as slash and burn, shifting cultivation, or 'rab' (wood ash) cultivation degrade forests.

3) Degradation of agricultural lands: Globally 5 to 7 million hectares of farmland is degraded each year. Loss of nutrients and overuse of agricultural chemicals are major factors in land degradation. Water scarcity is an important aspect of poor agricultural outputs. Salinization and water logging has affected a large amount of agricultural land worldwide.

4) Our fertile soils are being exploited faster than they can recuperate.

5) Forests, grasslands and wetlands have been converted to agricultural use, which has led to serious ecological questions.

6) Use of genetically modified seed variety, without minding the conducive environment for such experimentation, will seriously affect the land ecosystem.

7) Our fish resources, both marine and inland, show evidence of exhaustion.

8) There are great disparities in the availability of nutritious food. Some communities such as tribal people still face serious food problems leading to malnutrition especially among women and children.

9) Loss of Genetic Diversity: Modern agricultural practices have resulted in a serious loss of genetic variability of crops. India's distinctive traditional varieties of rice alone are said to have numbered between 30 and 50 thousand. Most of these have been lost to the farmer during the last few decades as multinational seed companies push a few commercial types. This creates a risk to our food security, as farmers can loose all their produce due to a rapidly spreading disease. A cereal that has multiple varieties growing in different locations does not permit the rapid spread of a disease.

Food security:

It is the ability of all people at all times to access enough food for an active and healthy life. It is estimated that 18 million people worldwide, most of whom are children, die each year due to starvation or malnutrition, and many others suffer a variety of dietary deficiencies. The earth can only supply a limited amount of food. If the world's carrying capacity to produce food cannot meet the needs of a growing population, anarchy and conflict will follow.

The following 3 conditions must be fulfilled to ensure food security

- Food must be available
- Each person must have access to it.
- The food utilized must fulfill nutritional requirements

Food security is closely linked with population control through the family welfare program. It is also linked to the availability of water for farming. Food security is only possible if food is equitably distributed to all. Many of us waste a large amount of food carelessly. This eventually places great stress on our environmental resources.

1) Institutional support for small farmers: A major concern is the support needed for small farmers so that they remain farmers rather than shifting to urban centers as unskilled industrial workers.

3) Protecting genetic diversity: The most economical way to prevent loss of genetic diversity is by expanding the network and coverage of our Protected Areas.

5) Environmental friendly farming methods: Shift from chemical agriculture to organic farming, practicing integrated nutrient management (INM), integrated pest management (IPM).

6) Several crops can be grown in urban settings, including vegetables and fruit which can be grown on waste household water and fertilizers from vermi-composting pits.

7) Prevention of water and land degradation: Pollution of water sources, land degradation and desertification must be rapidly reversed.

8) Population control: Most importantly food supply is closely linked to the effectiveness of population control programs worldwide.

9) Education: Educating women about nutrition, who are more closely involved with feeding the family, is an important aspect of supporting the food needs/security of many developing countries.

10) Changing food habits : Today the world is seeing a changing trend in dietary habits.

MODERN AGRICULTURE

Modern agriculture makes use of hybrid seeds of single crop variety, technologically advanced equipment, fertilizers, pesticides and water to produce large amounts of single crop.

Problems using fertilizers

1. **Micronutrient imbalance:** Chemical fertilizers used in modern agriculture contain Nitrogen, Phosphorus and Potassium (N,P,K) which are macronutrients. Excess use of fertilizers in fields causes micronutrient imbalance. Ex: Excessive use of fertilizers in Punjab and Haryana caused deficiency of micronutrient Zinc thereby affecting productivity of soil.
2. **Nitrate pollution**: Excess Nitrogenous fertilizers applied in fields leach deep into the soil contaminating the groundwater. If the concentration of nitrate in drinking water exceeds 25 mg/L it leads to a fatal condition in new-born babies. This condition is termed "Blue Baby Syndrome"
3. **Eutrophication**: The application of excess fertilizers in fields leads to wash off of the nutrient loaded water into nearby lakes causing over-nourishment. This is called "Eutrophication". Eutrophication causes the lakes to be attacked by "algal blooms". Algal blooms use nutrients rapidly and grow fast. Their life is short, they die and pollute water thereby affecting aquatic life in the lake.

Problems in using Pesticides:

In order to improve crop yield, pesticides are used indiscriminately in agriculture.
Pesticides are of two types:

1. First generation pesticides that use Sulphur, Arsenic, Lead or Mercury to kill pests
2. Second generation pesticides such as Dichloro Diphenyl Trichloroethane (DDT) used to kill pests. These pesticides are organic in nature. Although these pesticides protect our crops from severe losses due to pests, they have several side-effects as listed below:

- **Death of non-target organisms**: Several insecticides kill not only the target species but also several beneficial not target organisms
- **Pesticide resistance:** Some pests that survive the pesticide generate highly resistant generations that are immune to all kinds of pesticides. These pests are called "superpests"
- **Bio-magnification:** Most pesticides are non-biodegradable and accumulis ate in the food chain. This is called bio-accumulation or bio-magnification. These pesticides in a bio-magnified form are harmful to human beings.

- **Risk of cancer**: Pesticide enhances the risk of cancer in two ways (i) It acts as a carcinogen and (ii) It indirectly suppresses the immune system.

WATER LOGGING

If water stands on land for most of the year, it is called water logging.
In water logged conditions, pore-voids in the soil get filled with water and soil-air gets depleted. In such a condition the roots of plants do not get enough air for respiration. Water logging also leads to low mechanical strength of soil and low crop yield.

CAUSES OF WATER LOGGING

1. Excessive water supply to the croplands
2. Heavy rain
3. Poor drainage

MEASURES TO PREVENT WATER LOGGING

1. Avoid and prevent excessive irrigation
2. Sub-surface drainage technology
3. Bio-drainage by trees like Eucalyptus

SALINITY

Water not absorbed by soil, is evaporated leaving behind a thin layer of dissolved salts in the top soil. This is called salinity of the soil. Saline soils are characterized by accumulation of soluble salts like sodium chloride, calcium chloride, magnesium chloride, sodium sulphate, sodium carbonate and sodium bicarbonates. Saline conditions are exhibited when pH is greater than 8.0

PROBLEMS IN SALINITY

1. Saline soils yield less crop

In order to remedy the condition of saline soils the following two techniques may be used:

1. Salt deposit is removed by flushing with good quality water
2. By using a sub-surface drainage system, the salt water is flushed out slowly.

CASE STUDIES

Canal irrigation in Haryana resulted in rising water table followed by water logging and salinity causing low crop productivity thereby huge economic losses.
Similarly the "Indira Gandhi Canal Project" in Rajasthan converted a big area into a "water soaked waste land".

In Delhi, accumulation of pesticides and DDT in the body of mothers caused premature deliveries or low birth weight infants.

Food centre at Center for Science and Environment (CSE) India reported Pepsi and Coca-Cola companies sold soft drinks with a pesticide content 30-40 times higher than EU guidelines permit. At the reported concentrations the pesticides damage the nervous system.

Lect.7 Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources. Case studies.

Energy is defined as 'the capacity to do work'. **Sun** is the primary source of energy. **Joule** is the standard unit of energy in **SI units**. Energy utilization is an index of economic development, which does not take into account of ill effects/damage on to environment.

Energy is available on earth in a number of forms and some forms may be used immediately while others might require some transformation. It is difficult to imagine Life without energy. All the developmental activities in the world are directly or indirectly dependent on energy. Both energy production and energy utilization indicate a country's progress.

Growing energy needs

Energy is essential to the existence of mankind. All industrial processes like mining, transport, lighting, heating and cooling in buildings need energy. With the growing population, the world is facing an energy deficit. Lifestyle change from simple to a complex and luxurious lifestyle adds to this energy deficit. Almost 95% of commercial energy is available from fossil fuels like coal and natural gas. These fossil fuels will not last for more than a few years.Hence, we must explore alternative fuel/energy options

RENEWABL„EENERGY„SOURCES:

Renewable energy sources are natural resources that can be regenerated continuously and are inexhaustible. They can be repeatedly used

1.SOLAR ENERGY:

The energy that we get directly from the sun is called solar energy. Nuclear fusion occurring in the sun releases enormous amount of energy in the form of heat and light. Several techniques are available for collecting, converting and using solar energy.

2.WIND ENERGY:

Wind is defined as moving air. Energy recovered from the force of wind is called wind energy. Wind energy is harnessed by the use of wind mills.

1. **Wind mills**: The force of blowing wind strikes the blades of the wind mill thereby causing it to rotate continuously. This rotational energy of the blades is used to drive several machines like water pump, flour mill and electric generators.
2. **Wind farms**: Several wind mills joined together in a definite pattern forms a wind farm. Wind farms generate large amounts of electricity.
3.

CONDITION The minimum speed required for satisfactory working of a wind generator is 15 kmph

ADVANTAGES:

1. It does not cause any air pollution
2. It is very cheap

3.OCEAN ENERGY
Ocean can be used for generating electricity:

1. **Tidal energy**: Ocean tides produced by virtue of gravitational force of sun and moon possess enormous amounts of energy. Tidal energy can be harnessed by constructing a tidal barrage.

2. **Ocean thermal energy;** There is a large temperature difference between surface level and deep water level of tropical oceans. This temperature difference can be utilized to generate electricity. This energy is called ocean thermal energy.

4.BIOMASS ENERGY
Biomass is organic matter produced by plants or animals. It is used as a source of energy. Biomass is generally burnt for heating, cooling and industrial purposes.

Ex: wood, crop residues, seeds, cattle dung, sewage, agricultural wastes, etc.

Biomass may be converted into energy in any of the following types:

1. **Biogas**: Biogas is a mixture of gases such as methane, carbondioxide, hydrogen sulphide, etc. It contains about 65% of methane gas as a major constituent. Biogas is obtained by the anaerobic fermentation of animal dung or plant wastes in the presence of water.

2. **Biofuels**: Biofuels are the fuels, obtained by the fermentation of biomass. Examples are Ethanol, Methanol.

3. **Hydrogen fuel**: Hydrogen can be produced by thermal dissociation or photolysis or electrolysis of water. It possesses high calorific value. It is non polluting as the product of combustion is water.

Non-renewable energy sources

1.Coal: Coal is a solid fossil fuel formed in several stages as burried remains of land plants that lived 300-400 million years ago and were subjected to intense heat and pressure over millions of years.

Various stages of coal: The various stages of coal during formation of coal from wood are:

i. Wood ii Peat iii Lignite iv Bituminousv Anthracite

Carbon content of anthracite is 90% and its calorific value is 8700 kcal. Carbon content of Bituminous coal, Lignite and Peat are 80%, 70% & 60% respectively. India has approximately 5% of the worlds coal. However, it is not of good quality as it has poor heat capacity.

	Renewable energy resources	Non-renewable energy resources

Disadvantages of using coal:

1. Burning coal produces carbondioxide which is the main cause for global warming
2. Coal contains impurities like Sulphur and Nitrogen which produce toxic gases when burnt.
3.

2. **Petroleum:** Petroleum or crude oil is a thick liquid consisting of more than 100 cumbustible hydrocarbons with small amounts of S, O and N as impurities. Fossil fuels are mainly sormed by the decomposition of dead plants and animals that were buried under lakes and oceans at a high temperature and pressure for millions of years. From the crude petroleum oil, various hydrocarbons are separated by fractional distillation of crude petroleum oil. At the present rate of usage, the world's crude oil reserves are expected to get over in the next 30 years.

LIQUIFIED PETROLEUM GAS (LPG)

The petroleum gas obtained during crackling and fractional distillation can be easily converted into liquid under high pressure as LPG. LPG is a colourless, odourless gas to which mercaptans are added to produce foul smell that aids in detection of LPG leaks.

NATURAL GAS Natural gas is found above the oil in oil well. It is a mixture of 50-90% methane and small amounts of other hydrocarbons. Its calorific value ranges between 12000 and 14000 kcal/m3.

1. Dry gas: Natural gas containing low hydrocarbons like d ethane, it is called dry gas.
2. Wet gas: Natural gas containing high hydrocarbons like propane and butane along with methane is called wet gas.

Natural gas is formed by decomposition of dead plants and animals buried under oceans at high temperature and pressure for millions of years.

NUCLEAR ENERGY:

Dr. Homi Bhabha was the father of nuclear power development in India. India has 10 nuclear reactors that produce 2% of India's electricity. Nuclear energy is produced by two types of reactions:

(i) **Nuclear fission:** Nuclear fission is a nuclear chain reaction in which the heavy nucleas is split into lighter nuclii by fast moving neutrons thereby releasing a large amount of energy.

Ex: Fission of Uranium235

(**ii) Nuclear fusion:** Nuclear fusion is a nuclear chain reaction in which lighter nucleus are combined together at extremely high temperatures to form heavy nucleus thereby releasing large amount of energy.

Ex: Fusion of Dueterium atoms to form helium with release of large amount of energy.

1.	It can be used again and again throughout its life.	It cannot be used again and again but one day it will be exhausted.
2.	These are the energy resources which cannot be exhausted.	They are the energy resources which can be exhausted one day.
3.	It has low carbon emission and hence environment friendly.	It has high carbon emission and hence not environment friendly.
4.	It is present in unlimited quantity.	It is present in limited quantity and vanishes one day
5.	Cost is low.	Cost is high.
6.	Renewable energy resources are pollution free.	The non-renewable energy resources are not pollution free.
7.	Life of resources is infinite.	Life of resources is finite and vanishes one day.
8.	It has high maintenance cost.	It has low maintenance cost as compared with the renewable energy resources.
9.	Large land area is required for the installation of its power plant.	Less land area is required for its power plantbinstallation.
10.	Solar energy, wind energy, tidal energy etc are the examples of renewable resources.	Coal, petroleum, natural gases are the examples of non-renewable resources

Lect 8 .Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification

Land is a major resource for, food production, animal husbandry, industry our growing human settlements, Forests, wild life and biodiversity. Land on earth is as finite as any of our other natural resources. Scientists today believe that at least 10 percent of land and water bodies of each ecosystem must be kept as wilderness for the long term needs of protecting nature and natural resources.

Soil types are red soil, black cotton soil, literate soil, alluvial soil, desert soil etc. In nature India is moving North East @5cm/yr (fastest continent) so the Eurasian plate deforms and India compresses by 4mm/year

LAND DEGRADATION:
The surface layer of land is called soil.
Fertility or productive capacity of the soil depends on the minerals it contains.
Minerals are mainly available to the top layer of the soil. Hence, the top layer is the best for vegetation.

Land degradation refers to deforestation or deterioration or loss of fertility or productive capacity of soil. The factors contributing to land degradation are listed below and discussed subsequently.

Soil erosion
Soil pollution
Salination and water logging
Shifting cultivation
Desertification
Urbanisation

Land degradation: It is the decline in land quality or reduction in its productivity or production potential caused by human activities. World wide 5 -7 m ha farm land is being degraded annually.

Mechanisms that initiate land degradation include

Physical processes**:** decline in soil structure leading to crusting, compaction, erosion, decertification, Ana vision, environmental pollution and unsustainable use of natural resources.

Chemical processes: Acidification, leaching, decrease in caitions retention capacity and loss of nutrients.

Biological processes: Reduction in total and biomass carbon and decline in land biodiversity.

Causes for land degradation:

i. Intensive irrigation leads to water logging and salinisation, on which crops cannot grow.

ii. The use of more and more chemical fertilizers poisons the soil so that eventually the land becomes unproductive.

iii. The roots of trees and grasses bind the soil. If forests are depleted, or grasslands overgrazed, the land becomes unproductive and wasteland is formed

iv. Land is also converted into a non-renewable resource when highly toxic industrial and nuclear wastes are dumped on it.

v. As urban centers grow and industrial expansion occurs, the agricultural land and forests shrink. This is a serious loss and has long term ill effects on human civilization.

vi. Land degradation/soil erosion due to deforestation is more evident on steep hill slopes in the Himalayas and in the Western Ghats. These areas are called '**ecologically sensitive areas' or ESAs**. To prevent the loss of millions of tons of valuable soil every year, it is essential to preserve what remains of our natural forest cover.

vii. vii. The rate of mangrove loss is significantly higher than the loss of any other types of forests. If deforestation of mangroves continues, it can lead to severe losses of biodiversity and livelihoods, in addition to salt intrusion in coastal areas and siltation of coral reefs, ports and shipping lanes.

Land use planning:

Land use planning is an iterative process based on the dialogue amongst all state holders aiming at the negotiation and decision for a sustainable form of land use. Land use planning creates the prerequisite required to achieve a type of land use, which is sustainable, socially and environmentally compatible, socially desirable and economically sound. Planning approaches often fail because global models and implementation strategies are applied and taken over automatically and uncritically.

Land use planning is not a standardized procedure which is uniform in its application world wide its content is based on an initial regional or local situation analysis. Land use planning should consider following principles.

1. It should take into account traditional strategies and local environmental knowledge.
2. Differentiation of state holders and the gender approach are core principles in land use planning.
3. The ecological, economic technical financial, social and cultural dimension of land use makes it necessary to work with inter disciplinary approach.
4. It should aim at finding solutions for present problems (soil erosion, low yield, and low income in rural house holds) with the planning towards long conservations and sustainable use of land resources.

Soil erosion:
is the loss or removal of the superficial layer of soil by the action of water, wind or human activities. **Factors influencing the extent of soil erosion are:**

Distribution, intensity and amount of rainfall:
Unequal distribution of rainfall results in heavy rainfall being restricted to a few months. The soil unable to absorb this heavy rainfall causes run-off water that removes layers of soil as it moves, resulting in soil erosion.

Slope of the ground: Steep slopes cause decreased infiltration and increased run-off resulting in more soil erosion.

Nature of the soil: Light, open soils lose more silt than heavier soils (loam) that swell-up by wetting.
Vegetation cover: Vegetation holds the soil in place by forming a network of roots of plants. Rainfall on thick vegetation causes negligible soil erosion. Rain falling on bare land causes soil erosion as top soil is loose.

Soil mismanagement: The following techniques listed below contribute to soil mismanagement:
Faulty methods of soil drainage
Overgrazing
Wrong methods of cultivation
Forest fires and
Removal of forest litter are common practices that aggravate soil erosion.
Erosion, floods and sedimentation result in deposition of silt and consequent clogging of irrigation canals.

Soil pollution: Soil pollution is defined as the reduction in productivity of soil due to presence of soil pollutants.

LAND AS A RESOURCE

Human and natural activities need space for their location and development. This space is provided by land which is put to various uses like food and energy production, waste-disposal, industrial, commercial and residential purposes.

Land houses the living species, water resources and raw material resources (minerals and ores).
Pattern of land use on earth is:
Arable land
Land for pastures and meadows
Forest land
Urban land and
Non-agricultural land

Desertification: It is land degradation occurring in arid, semiarid and dry subhumid areas of the world. It is a process where in fertile lands become arid through land mismanagement or climate changes. Many deserts in the world are man-made.

Desertification is taking place much faster worldwide than historically and usually arises from the demands of increased populations that settle on the land in order to grow crops and graze animals.

Causes of desertification:

1) ***Overgrazing*:** By pounding the soil with their hooves, livestock compact the substrate, increase the proportion of fine material, and reduce the percolation rate of the soil, thus encouraging erosion by wind and water. Grazing and the collection of firewood reduce or eliminate plants that help to bind the soil.

2) ***Increased population*:** Livestock pressure on marginal lands accelerates desertification.

3) ***Deforestation practices***: Loss of vegetation results in surface run off as there are no plants to bind the soil and resulting in soil erosion and depletion of nutrients.

4) Increased food production from marginal lands in arid or semi- arid areas.

5) Irrigation projects in areas with no drainage facility.

6) Shifting of sand dunes by wind storms

Effects: A major impact of desertification is biodiversity loss, and loss of productive capacity, such as the transition from grassland dominated by perennial grasses to one dominated by perennial shrubs. In extreme cases, it leads to the destruction of lands' ability to support life.

Control of desertification

1. Afforestation and planting of soil binding grasses can check soil erosion, floods and water logging.

2. Crop rotation and mixed cropping improve the fertility of the soil. It would increase production which can sustain large population.

3. Desertification can be checked by artificial bunds or covering the area with proper type of vegetation.

4. Shifting of sand can be controlled by mulching (use of artificial protective covering.)

4. Salinity of the soil can be checked by improved drainage. Saline soil can be recovered by leaching with more water, particularly where water table of the ground is not very high.

ROLE OF AN INDIVIDUAL IN CONSERVATION OF NATURAL RESOURCES

Until fairly recently mankind acted as if he could go on for ever exploiting the ecosystems and natural resources such as soil, water, forests and grasslands on the earth's surface and extracting minerals and fossil fuels from underground.
But, in the last few decades, it has become increasingly evident that the global ecosystem has the capacity to sustain only a limited level of utilization. Biological systems cannot go on replenishing resources if they are overused or misused.
At a critical point, increasing pressure destabilizes their natural balance. Even biological resources traditionally classified as 'renewable' - such as those from our oceans, forests, grasslands and wetlands, are being degraded by overuse and may be permanently destroyed.

And no natural resource is limitless. 'Non-renewable' resources will be rapidly exhausted if we continue to use them as intensively as at present.

The two most damaging factors leading to the current rapid depletion of all forms of natural resources are increasing 'consumerism' on the part of the affluent sections of society, and 'rapid population growth'. Both factors are the results of choices we make as individuals.

Energy conservation
Turn off lights and fans as soon as you leave the room.
Use tube lights and energy efficient bulbs that save energy rather than bulbs. A 40- watt tube light gives as much light as a 100 watt bulb.
Keep the bulbs and tubes clean. Dust on tubes and bulbs decreases lighting levels by 20 to 30 percent.
Switch off the television or radio as soon as the program of interest is over.
A pressure cooker can save up to 75 percent of energy required for cooking. It is also faster.
Keeping the vessel covered with a lid during cooking, helps to cook faster, thus saving energy.

Water conservation:
· Keep taps closed when brushing teeth and taking a bath
· Use drip and sprinkler type of irrigation in agricultural fields
· Practice rain water harvesting techniques
· Reuse the waste water from kitchens and bath for garden use

Soil conservation:
· Do not cut trees and induce soil erosion
· Practice contour farming, agro forestry and strip cropping
· Practice no till farming for less soil disturbance
· Avoid over use of fertilizers, pesticides and water logged conditions
· Use organic fertilizers and vermicompost
· Practice integrated pest management practices

Lect.9 Ecosystems: Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem